

code :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int low, mid, high, n, m;
```

```
    int a[50], b[50];
```

```
    printf("Enter the size of arrays: ");  
    scanf("%d %d", &n, &m);
```

```
    printf("Enter array 1 elements:");  
    for (int i=0; i<n; i++)  
        scanf("%d", &a[i]);
```

```
    printf("Enter array 2 elements:");  
    for (int j=0; j<m; j++)  
        scanf("%d", &b[j]);
```

```
    mergeSort (m, n, a, b);
```

```
}
```

```
void mergeSort (int m, int n, int a[], int b[])
```

```
{
```

```
    int i=0, j=0, c[50];
```

```
    while (i<m & j<n)  
    {
```

```
        if (a[i] < b[j])  
        {  
            c[k] = a[i];  
            i++;  
            k++;
```

```
    }
```

```

else
{
    c[k] = b[j];
    j++;
    k++;
}
}

while (i < m)
{
    c[k] = a[i];
    i++;
    k++;
}

while (j < n)
{
    c[k] = b[j];
    j++;
    k++;
}

printf("Merged array : ");
for (int i = 0; i < (m+n); i++)
    printf("%d ", c[i]);

return;

```

~

Time :

$T(n) = O(m+n) + O(m) + O(m) + O(m+n)$

Output: $m=10$ $n=10$

Enter array 1 elements : 1 3 5 7 9 11 13 15 17 19

Enter array 2 elements : 0 2 4 6 8 10 12 14 16 18

Merged array : 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19